4.6 Best practices for speaking into a two-way radio

Radios are used to improve safety. It's important that the messages you transmit are clearly heard and understood.

- If possible, hold the microphone directly in front of your mouth, just 1 to 2 inches away. This helps minimize surrounding noises.
- Speak directly into the microphone, rather than across it. Radios are designed to be spoken directly into.
- As much as possible, shield the microphone from surrounding noises.
- Position radios and microphones away from car radios or other noisy equipment.
- Speak clearly and at a normal pace; neither too fast nor too slow. And if someone on the other end is hard to understand, don't be afraid to say so.
- In congested areas with lots of traffic, it's especially important to use proper radio calling procedures, meaning use previously established call signs.

4.7 Radio use on resource roads

When radio use is required for travel on resource roads, *these guidelines* from the BC Ministry of Forests, Lands and Natural Resource Operations should be followed.

It is very important that users have an accurate map of the resource road channel assignments for the resource roads they will be travelling on. There is *channel assignment information* on the Innovation, Science, and Economic Development Canada website.

Section 5: Traffic Control (Flagging) Practices

Pilot car operators are often required to conduct traffic control (flagging) as part of an oversize load move. Any pilot car operator that is conducting flagging must be a certified traffic control person. In each situation, a pilot car operator will have to decide if it is best to flag from within their vehicle or outside their vehicle.

> Illuminated traffic stop paddles, with LEDs around the perimeter of the sign, are an acceptable enhancement as per *Work Safe Occupational Health and Safety Regulation Guidelines Part 18, Section 9a*.

5.1 Flagging from Inside a Pilot Car

The following are conditions where it may be appropriate to control traffic from within a pilot car. These are straightforward, simple and brief situations. An example of a simple flagging situation where traffic control may be conducted from within a vehicle would be a short closure at a two-lane bridge or tunnel where the load needs to straddle the centreline in order to pass.

- Daylight hours only
- Short duration (15 minutes or less)
- Good sight distances for approaching vehicles in both directions
- Work on a two-lane highway
- No adjacent or nearby intersections or accesses
- Low traffic volumes
- Traffic control only needs to direct to stop or proceed

Flagging from inside a Pilot Car requires the following equipment:

- Retro-reflective hand held stop/slow C-027 paddle
- *Optional instead of hand help stop/slow sign:* Vehicle-mounted stop R-001 sign as per *Section 4.2.1*
- The appropriate Class 2 safety apparel as per *Section 5 of the Traffic Management Manual for Work on Roadways*

Pilot cars which use a dynamic message sign may control traffic from inside the pilot car at night, under the above circumstances.

5.2 Flagging Outside a Pilot Car

When conducting traffic control from outside a vehicle, refer to the *Section 5 of the Traffic Management Manual for Work on Roadways* for information on positioning, signals, and safety requirements. Remember to always have an escape route and be aware of your surroundings.

Except for situations described in *Section 5.1*, traffic control should be performed from outside the pilot car. This includes:

- Longer duration
- Sight distance limitations
- Multi-lane highways
- Near intersections, or there are intersections/accesses within the area that is being closed
- Higher traffic volumes
- Where traffic may need to be guided or directed to perform an action other than stopping.

Flagging from outside a Pilot Car requires the following equipment:

- Retro-reflective stop/slow C-027 paddle, hand-held or pole mounted
- Illuminated baton with light that appears red or flashlight fitted with a red signalling wand
- The appropriate Class 3 safety apparel as per *Section 5 of the Traffic Management Manual for Work on Roadways*
- Two-way radios to maintain communication with the load and other pilot vehicles.

In all cases, a TCP must consider their visibility when choosing a location from which to stop traffic. It is important to remember that the distance required for an approaching vehicle to stop increases with approach speed, on wet roads, or on downgrades. Drivers need enough space to see the TCP, react to their sign, and come to a stop. *Table 5.1 – Stopping Sight Distance* can be used as a guide and provides a summary of stopping sight distance for passenger vehicles on a flat, wet roadway.

Table 5.1 – Stopping Sight Distance

Speed (km/h)	Stopping Sight Distance (m)
60	85
70	110
80	140
90	170
100	210
110	250
120	290

When standing near the pilot vehicle, the headlights and other accessory lights can draw attention away from the TCP and obscure their visibility. The pilot vehicle should be oriented so that it is not shining lights directly behind the TCP (when viewed by approaching vehicles). This arrangement can make it difficult for approaching drivers to see the TCP over the headlight glare. In this situation, the headlights on the pilot vehicle should be switched to parking lights so that the vehicle itself is still visible, but does not drown out the TCP.

Note: It is important to use your judgment as terrain or circumstances may require additional measures. If you are standing outside your vehicle flagging for less than 15 minutes, you don't always need signs and other devices. However, planning an escape route, being prepared to use it, and positioning yourself where you can see and be seen is extremely important. See *Section 4.3* for more on signs and devices for traffic control.

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Section 6: Pilot Car Load Movement - General

6.1 Pilot Car Operation for Different Load Characteristics

Each oversize load is different. It may be overwidth, overheight, overlength, and/or overweight, and those characteristics help to establish what the pilot car operator needs to watch for and be aware of during the move.

> **Note**: If it is necessary to take actions that will affect the movement of other traffic around the load, those actions must be taken by a certified traffic control person or peace officer.

6.1.1 Overwidth Loads

In addition to being a moving warning device to let other drivers know that something big may be coming into their lane, pilot car operators should watch for areas where the road narrows, shoulder signs or rock outcroppings that may be risks to the load, spots where tunnel or bridge edges will require extra caution, curbs or concrete barriers that may require ground clearance, etc. Also, keep the load driver informed about obstacles such as cyclists or stopped vehicles ahead on the road shoulder.

For a lead pilot car, it may help to position your vehicle close to the centre line.

6.1.2 Overheight Loads

The physical height of an overheight load must match or be lower than the height indicated in the permit. Watch for bridges, overpasses and other obstacles that should be communicated to the load driver.

Power lines frequently cross highways, and only professionals with the utility companies, or their contractors, can handle their lines. *Part 19 of the Occupational Health and Safety Regulation* is a good source of information about maintaining a safe distance from lines.

See the Trip Planning section for more information about checking height clearance limits on your route in advance, and the Common Constriction Points section for more information about travelling through overhead obstacles.

6.1.3 Overlength Loads

Curves and turns onto and off of roadways are key for overlength loads. The load driver may need to swing very wide in order to complete a turn without the middle part of the load posing a danger to the inside corner and any pedestrians, cyclists or road infrastructure located there. When negotiating tight curves to the left, the middle part of the load may encroach into the next lane. In either direction, the tail swing may need careful control.

It may take extra time for the load to clear an intersection or rail crossing. Also, if a long vehicle combination has low ground clearance, be aware of and careful at rail crossings, since the load must clear the rails.

6.1.4 Overweight Loads

Many permits for overweight loads have specific conditions for crossing bridges along the route. It's important to know the locations of the bridges in advance, together with the exact conditions for the crossing. Common conditions include:

- Travel down the centre line, or straddling two lanes
- Maximum speed of 10 km/h
- Must be the only vehicle on the bridge while crossing

The permit documents will tell you which conditions apply at each bridge. Often bridge crossing conditions make it necessary to hold traffic while the load crosses. See *Section 8*: Pilot Car Load Movement Layouts - Structures for further guidance.

Term weight permits have a permit attachment (*Form CVSE1011*) that lists bridges with weight restrictions on major provincial roads.

6.2 Positioning on the Highway

Basic Pilot Car positioning is set out in the Commercial Transport Regulation, Division 8 or for more complex situations, in the permit and accompanying documents. When road conditions dictate that the use of the pilot car in another position (or at a greater distance) would be more effective, the pilot car position may be temporarily reassigned. For example: A pilot car is assigned to the rear of an overlength load on a two-lane highway. The load is about to enter a highway segment that has curves significant enough to cause the load to encroach on the oncoming lane of traffic. The pilot car may be temporarily reassigned to the front to warn oncoming traffic.

When escorting a load, the Pilot Car shall have its sign displayed, headlights on and amber flashing lights on.

6.3 Passing a Slower Vehicle

Although it is rare for an oversize load to overtake a slower moving vehicle, it may occur. Be cautious and aware of how much time you will need to spend in the passing lane. A decision to pass will be made by the load driver, and the pilot cars will keep the load driver informed of their movements and movements by other traffic in the area.

If there is a lead pilot car, one of its roles is to inform the other vehicles about oncoming traffic, in case they need to move to the right for safety. Remember that the rear pilot car's forward vision will be blocked by the load. They are relying on communication from the forward-most vehicle to tell them if they need to merge back into the right lane.

If there is a rear pilot car, its role is to move left just before the load moves left, blocking other vehicles from passing and informing the load driver when they are in position. As soon as it passes the slower vehicle, the rear pilot car driver notifies the load driver by radio that it is safe to move right, and then follows the load back into position in the right lane.

Figure 6.1A Initiating a passing manoeuvre – rear pilot car moves left

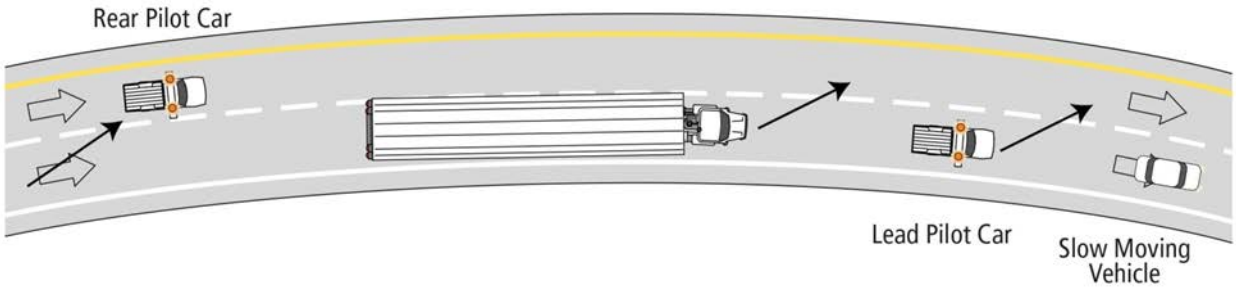


Figure 6.1B Truck moves left once the rear pilot car is in position

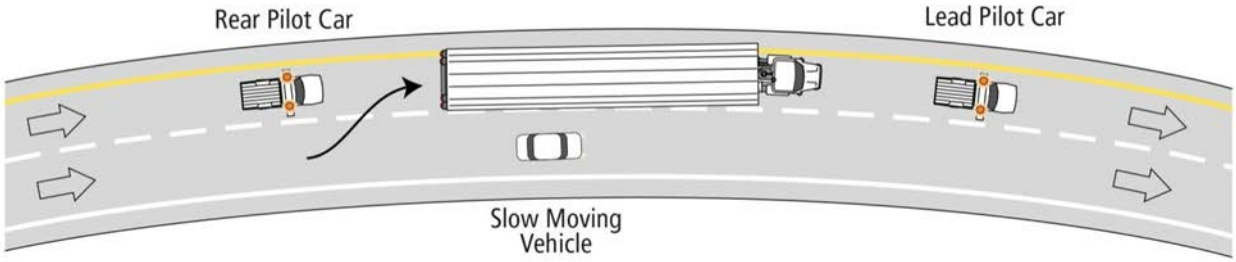
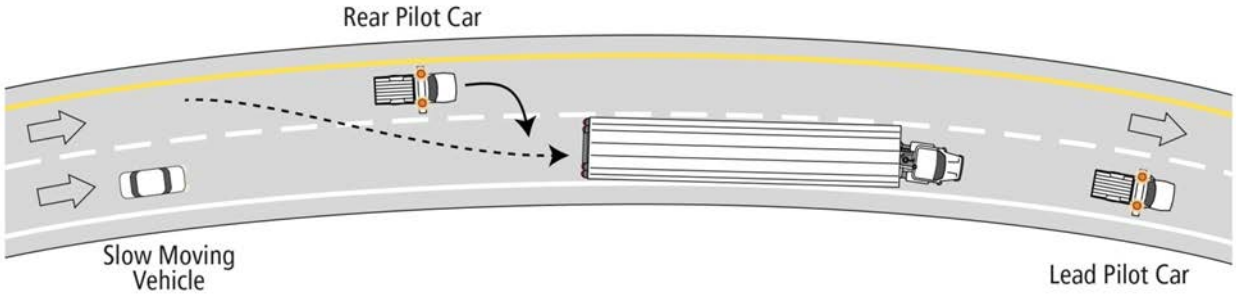


Figure 6.1C Rear pilot car notifies the load driver that the slow vehicle is now passed, and the load and rear pilot car move right



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Section 7: Pilot Car Load Movement Layouts – Roadway Type

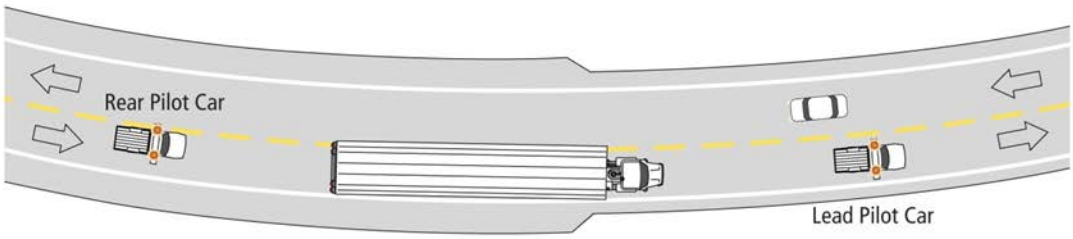
7.1 Two Lane Road Positioning – One or Two Pilot Cars

On two lane highways where only one pilot car is used, it typically occupies the lead pilot car position (except for loads that are overlength and not exceeding 3.2 m wide, when it travels to the rear.) The lead pilot car is watching for hazards that should be communicated to the load driver, including overhead restrictions, and making oncoming traffic aware of the approaching load. Where two pilot cars are required, one is positioned to the front and one to the rear of the load.

For overlength only, or when there is a rear pilot car because two cars were required by the permit, the rear pilot car travels behind the permitted load at a safe distance. The rear pilot car should drive as close to the left side of the lane as practicable without crossing the centreline.

Typically, the oversize load travels as far to the right as possible given the road geometry and obstacles.

Figure 7.1 Two lane road with two pilot cars



7.2 Two and three lane highway positioning – one or two pilot cars

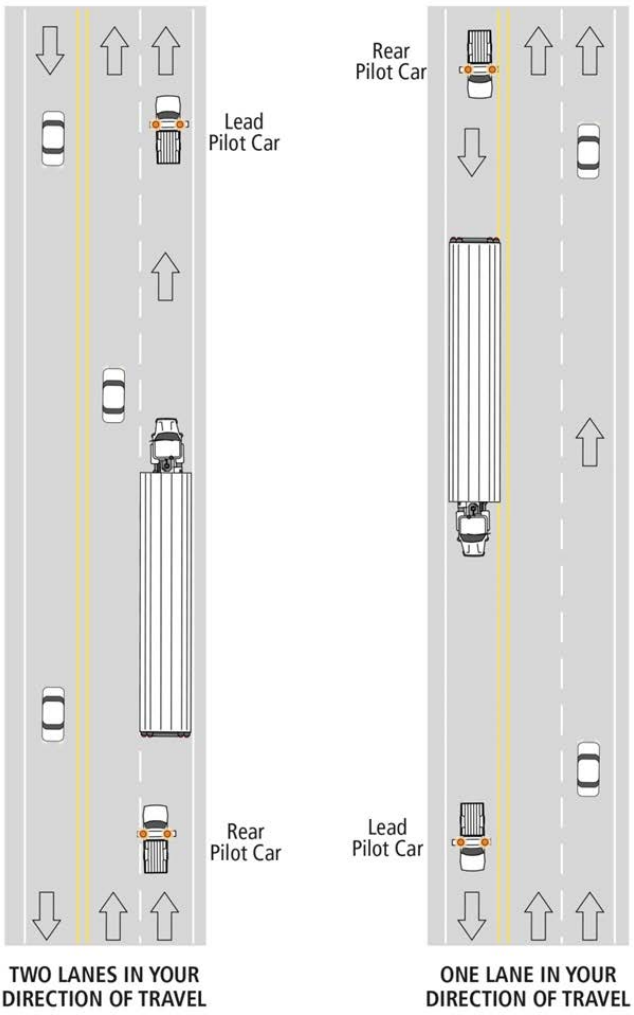
The lead pilot car is watching for hazards that should be communicated to the load driver, including overhead restrictions, and making oncoming traffic aware of the approaching load. The position of the rear pilot car may change depending on whether travelling uphill or downhill, the number of lanes in its direction of travel, and on the position of the oversize load.

With two lanes in your direction of travel, often on an uphill slope, if it is unsafe for vehicles to pass, the rear pilot car positions itself in the lane to the left of the load, blocking traffic approaching from the rear.

With one lane in your direction of travel, often on a downhill slope, the rear pilot car travels behind the permitted load at a safe distance. The rear pilot car should drive as close to the left side of the lane as practicable without crossing the centreline.

Typically, the oversize load travels as far to the right as possible given the road geometry and obstacles.

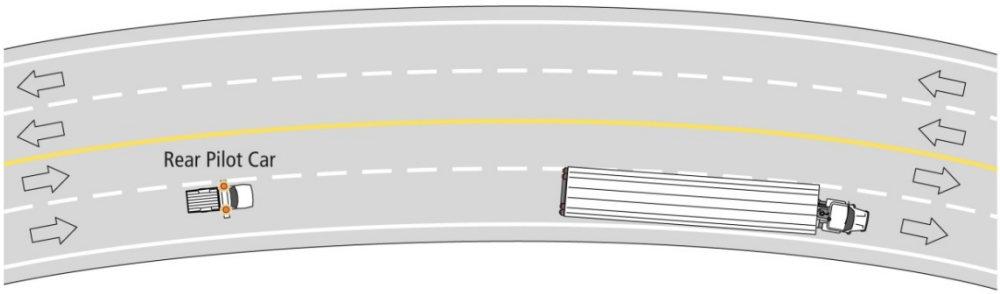
Figure 7.2 Passing or Climbing Lane – Two Pilot Cars



7.3 Multilane Highway – One and Two Pilot Cars

For highways with 4 or more lanes, or on divided highways with one pilot car, the pilot car is positioned in the rear pilot car position to alert drivers approaching the load from the rear. The pilot car typically operates 4 to 8 seconds behind the load.

Figure 7.3A Multilane highway - one pilot car



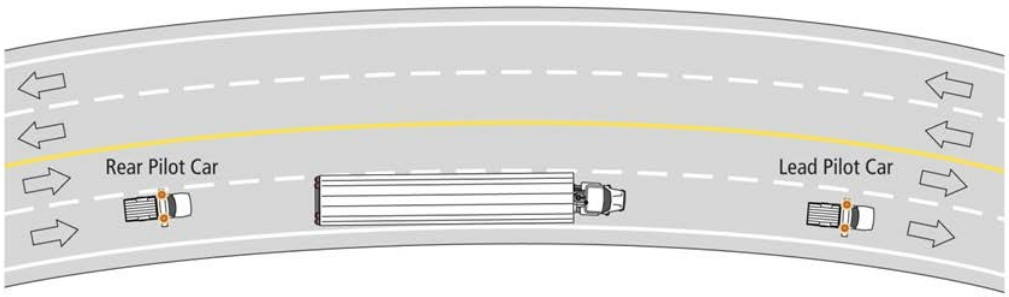
When two pilot cars are required, one pilot car travels to the front and one to the rear, in the same lane as the load.

The lead pilot car is watching for hazards that should be communicated to the load driver, including overhead restrictions, and making oncoming traffic aware of the approaching load.

The rear pilot car should drive as close to the left side of the lane as practicable without crossing the lane line, unless necessary to place the vehicle in line with the left side of the load to protect it from other traffic.

Typically, the oversize load travels as far to the right as possible given the road geometry and obstacles.

Figure 7.3B Multilane highway - two pilot cars



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Section 8: Pilot Car Load Movement Layouts – Structures

8.1 Two-lane bridges with two-way traffic – Two pilot cars

One common reason to close a bridge for a load to pass is the width of the load. Another is that weight permits often set out specific conditions for bridge crossings, which may necessitate closure of the bridge. In some cases, you may be able to allow traffic to the rear to follow the load across the bridge. Sometimes, though, the permit conditions will require that traffic be stopped in both directions while the load proceeds.

The diagrams in this section show two pilot cars. See *8.2 Two Lane bridge with two-way traffic – Three Pilot Cars* if your permit requires three pilot cars.

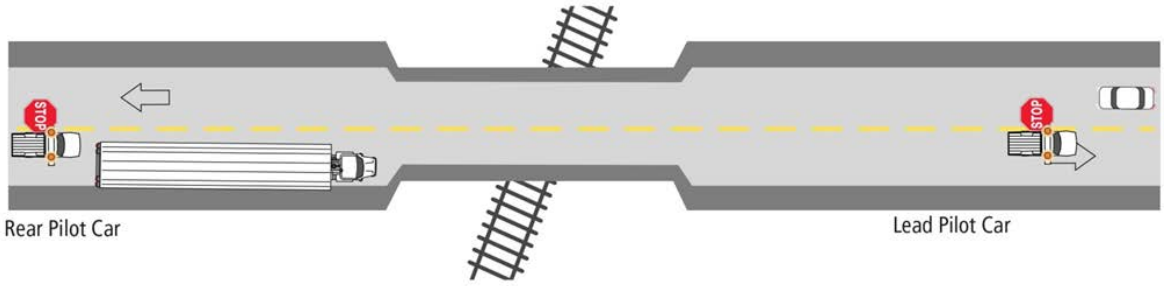
See *Section 5* for guidance about stopping distances and whether to flag from inside or outside the vehicle.

In this scenario, the lead pilot car may need to move well out in front of the load as the team approaches a bridge stop location, to the first location past the bridge that has appropriate sight lines for the oncoming traffic, and sufficient room between the lead pilot car and the bridge so that the load will be able to clear the bridge completely before the oncoming traffic is released. The lead pilot car will rely on communication with the load driver to know when the load is approaching the bridge. If it is necessary for the lead pilot car to wait before commencing the traffic stop, it should do so on the shoulder.

Once the lead pilot car has traffic stopped, it radios the rest of the team (with a description of the last car on its way to the load, if applicable), and the load proceeds across the bridge, with the rear car either trailing or holding traffic (if permit conditions require that no other traffic should be on the bridge with the load.)

As soon as the load clears the bridge, the rear pilot car radios that the lead pilot car can release traffic. The lead pilot car waits for the load and the rear pilot car to reach its location and the team proceeds.

Figure 8.1 Two lane bridge with two-way traffic – Load approaching the bridge, waiting for the lead pilot car to advise that traffic is stopped before proceeding



8.2 Two-lane bridges with two-way traffic – Three pilot cars

If your permit requires three pilot cars, you will normally have two in front (scout and lead) and one to the rear.

The scout pilot car will normally be moving well ahead of the rest of the vehicles in the move team, in order to stop traffic where sight lines are good, as needed for the load to proceed. This is no different in a two-lane bridge crossing with two-way traffic; oncoming traffic should be stopped at the first location past the bridge that has appropriate sight lines for the oncoming traffic, and sufficient room between the scout pilot car and the bridge so that the load will be able to clear the bridge completely before the oncoming traffic is released. The scout pilot car will rely on communication with the lead pilot car to know when the load is approaching the bridge. If it is necessary for the scout pilot car to wait before commencing the traffic stop, it should do so on the shoulder.

In this scenario, the lead pilot car is just ahead of the load as it travels to the bridge and, once the scout pilot car radios that traffic is stopped (with a description of the last car on its way to the load, if applicable), the lead pilot car waits for the last oncoming car and then proceeds to the scout pilot car's position and takes over there. The scout pilot car can then proceed to its next stop location. Some pilot car teams prefer to use a 'leap-frog' approach where the scout and lead pilot cars trade positions at each stop rather than have the lead pilot car take over from the scout car at each stop location.

Meanwhile, the load proceeds across the bridge, with the rear car either trailing or holding traffic (if permit conditions require that no other traffic should be on the bridge with the load.)

As soon as the load clears the bridge, the rear pilot car radios that the lead pilot car can release traffic. The lead pilot car waits for the load and the rear pilot car to reach its location and radios the scout pilot car that all vehicles are once more on the move.

> **Note:** When the scout or lead pilot car is holding oncoming traffic, if more than a few vehicles are queued, the pilot car operator may choose to ask the first stopped car to wait until the oversize load reaches them and then proceed. Once they have agreement from that first stopped driver, they can slowly travel down the line of cars, with their stop sign out, partly to inform the waiting drivers of the reason for the delay (if time allows), but importantly, to protect the last stopped vehicle by keeping the pilot car lights and stop sign at the point where new arriving vehicles must stop.

Figure 8.2A Two lane bridge with two-way traffic – Three Pilot Cars

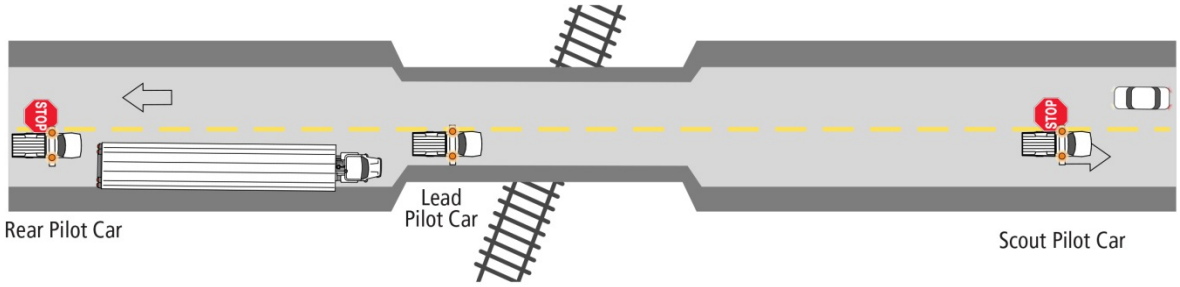
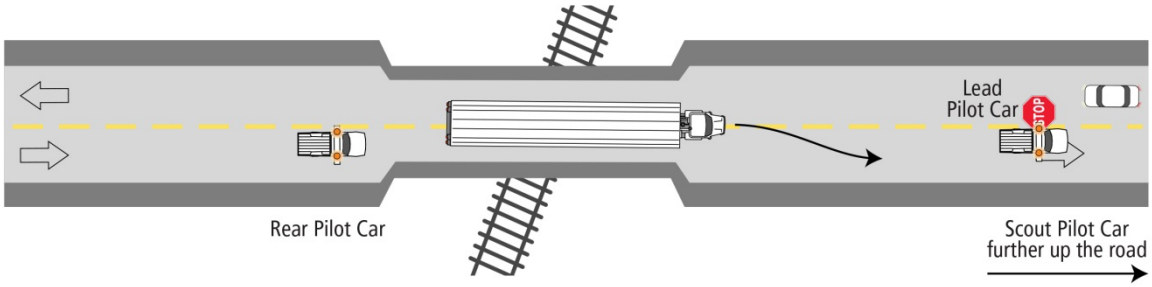


Figure 8.2B Two lane bridge with two-way traffic – Three Pilot Cars



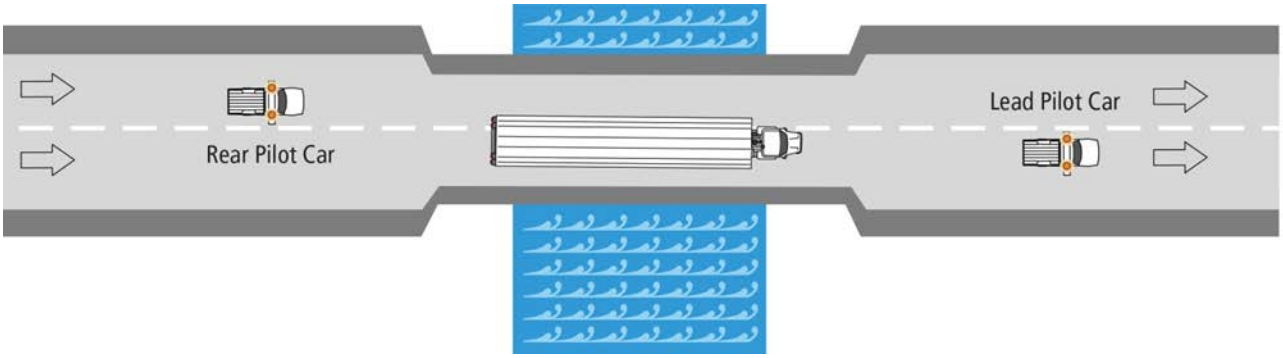
8.3 Multilane-bridges with one way traffic

If two pilot cars are required for the move, the lead pilot car should notify the load driver as far in advance as possible of any approaching obstructions.

If the load must cross straddling two lanes (or down the centre of the bridge), but there is no permit requirement that other traffic must be kept off the bridge during the crossing, the rear pilot car will move left just before the load moves left, blocking other vehicles from passing and informing the load driver when they are in position. As soon as the load and rear pilot car have crossed the bridge, the rear pilot car driver moves right into their normal travelling position and notifies the load driver by radio that it is safe to move right.

If permit conditions require that no other vehicles may be on the bridge with the load, the lead pilot car should join the rear pilot car behind the load in advance of the bridge, and together, they hold traffic while the load moves across. Depending on the terrain and the length of the bridge, it may be possible to perform this manoeuvre at rolling speed, but a full stop may be required.

Figure 8.3 Multilane bridges with one-way traffic



8.4 Tunnels and other restricted sight distance locations

Controlling traffic for the movement of oversized loads where sight distance restrictions exist is a challenging situation for pilot cars. The reason is the lack of advance knowledge for approaching motorists of where the oversize load is situated. At locations with restricted sight distance (tunnels or other blind corners), traffic control may be required, especially if the load may encroach into an oncoming lane.

Safety Planning: In situations where the route follows a winding road, a scout pilot car is beneficial. The scout pilot car can travel further ahead and provide advance notification to the load and other pilot cars of constraints as well as establish traffic control. If a scout pilot car is used, it can assume the traffic control and some of the communication duties of the lead pilot car as described below. For traffic control (flagging) information see Section 7.

If no scout pilot car is used, the lead pilot car should travel well ahead of the load, and communicate with the oversize load driver about constraints and any approaching traffic.

At any locations where the load will encroach into an oncoming lane and traffic control is required, the scout or lead pilot car will choose a location with good lines of sight for approaching traffic.

See *Section 5* for guidance about stopping distances and whether to flag from inside or outside the vehicle.

The lead pilot car will rely on communication with the load driver to know when the load is approaching the constrained area. If it is necessary for the lead pilot car to wait before commencing the traffic stop, it should do so on the shoulder.

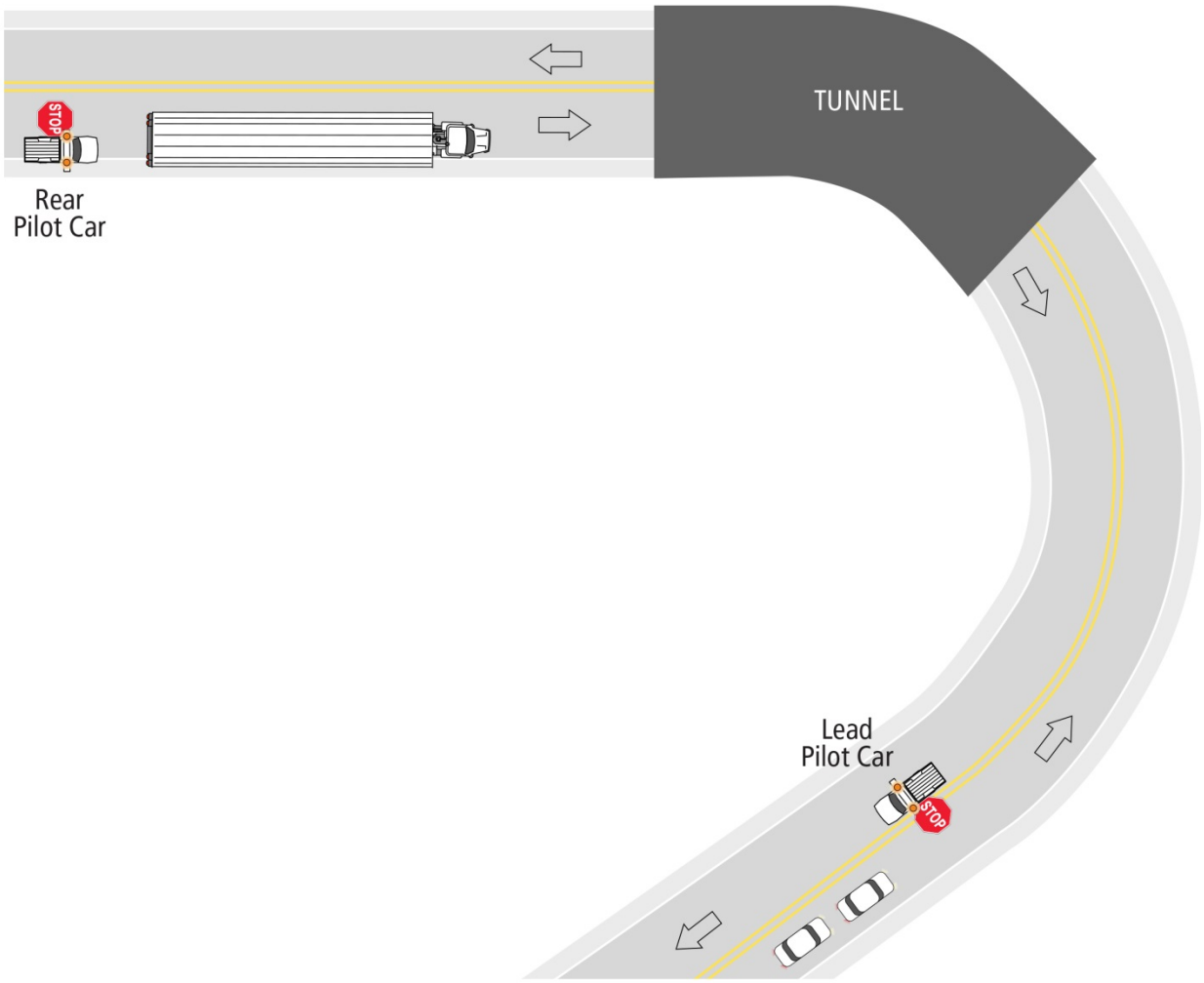
Once the lead pilot car has traffic stopped, it radios the rest of the team (with a description of the last car on its way to the load, if applicable), and the load proceeds through the constrained area, with the rear pilot car either trailing, or holding traffic to the rear, if necessary.

As soon as the load clears the tunnel or other restricted sight location, the rear pilot car radios that the lead pilot car can release traffic. The lead pilot car waits for the load and the rear pilot car to reach its location and the team proceeds.

On very curvy stretches of highway, it may be necessary to divide the highway into sections and perform traffic control for each section separately, moving the load from one safe stopping location to the next. In this case, a scout pilot car would be recommended, particularly if there are other access points to the highway between the scout pilot car and the load. The rear pilot car stops with the load, in line with the left edge of the load, and assists the load to re-enter the highway from each stopping location.

Where curves, or terrain features make radio communication challenging or impossible, the lead or scout pilot car stopping traffic may use the ‘baton method’, giving a baton or flag to the last car proceeding towards the load and asking the driver to pass that baton or flag to the rear pilot car.

Figure 8.4 Tunnels and other sight distance restricted locations



8.5 Railway Crossings

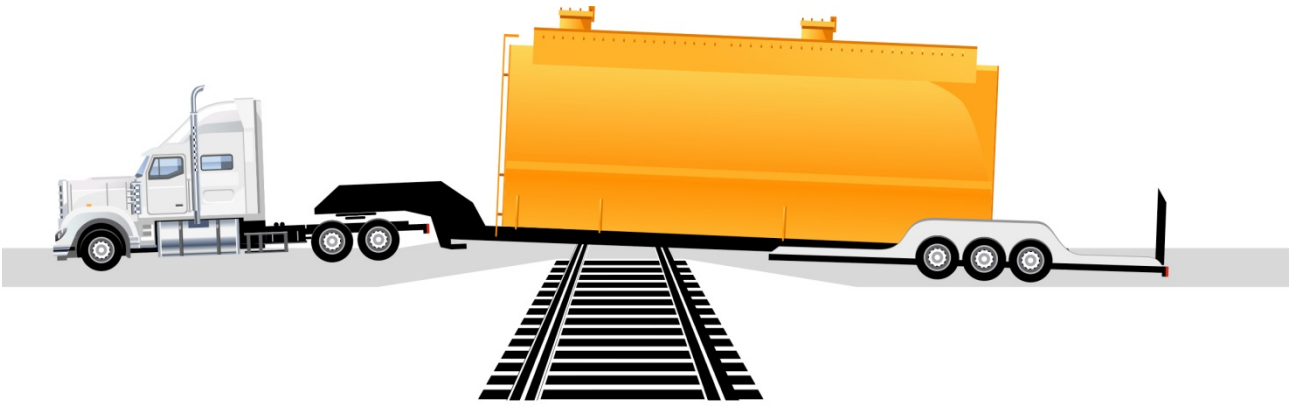
Overlength loads must also be evaluated for railway crossings to make sure that long loads do not:

- become ‘high centered’, as shown in Figure 8.5
- encounter problems resulting from the time it may take for a long load to clear a rail crossing, particularly near intersections or other road geometry features

Railway crossings should be assessed for:

- Amount of daily rail traffic to be encountered
- Crossing profiles between highway and rails
- Slope on the approaches and departures, the length of crossing, the number of tracks, and the road condition over the tracks both on approach and departure including curves and the need to make turns.
- In cases where railway crossings are a concern for the move, ensure that emergency contact numbers for the railways are available
- Advance contact should be made with the railroad if there is any doubt the load can safely negotiate the crossing

Figure 8.5 Railway crossing clearances



The profiles of railway crossings encountered along a route can vary, making them a potential location for a load to get hung up, or ‘high centred’. Always assess all railway crossings along a route for clearances, especially if “lowboy” trailers are involved in the move.

If you think the rails may have been damaged by a move, you should contact the railway immediately so that they can ensure safe rail operations are maintained.

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Section 9: Pilot Car Load Movement Layouts – Intersections

9.1 Driving in urban conditions

In British Columbia, primary highways often travel through the heart of municipalities and smaller communities. Movement through urban areas requires additional care and constant communication within the move team. Some additional issues to be considered include:

- An increase in the number and density of height clearance restrictions, such as low wires and signs;
- Pedestrians and cyclists, which may be difficult for the load driver to see;
- An increase in activity, including the number of other drivers, private accesses, unexpected obstacles;
- Raised channelization (eg. pedestrian islands) and landscape treatments;
- Increased traffic volumes building up behind the load, which may need traffic control management.
- Limited sightlines due to buildings and other structures.

Some urban obstacles may require that the load move temporarily into lanes meant for opposing traffic, or in some other manner that requires traffic control. Appropriate approvals must be in place.

The number of pilot cars required by a permit depends on the load characteristics, however, for loads in excess of 4.88 m high, the use of three pilot cars (scout, lead, and rear) is recommended.

Relatively smaller wide or long load moves may be conducted with one pilot car, but at least two are recommended in areas with tighter turns, higher traffic volumes or obstructions that require the load to travel in opposing lane(s).

Figure 9.1A Driving in urban conditions – Traffic entering intersection is stopped

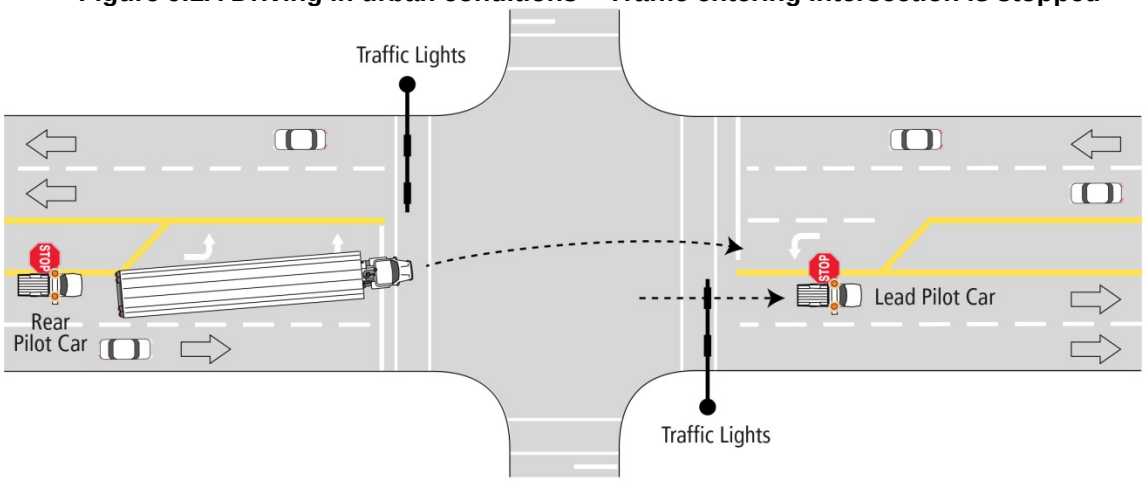


Figure 9.1B Driving in urban conditions – Load uses opposing lane to cross intersection

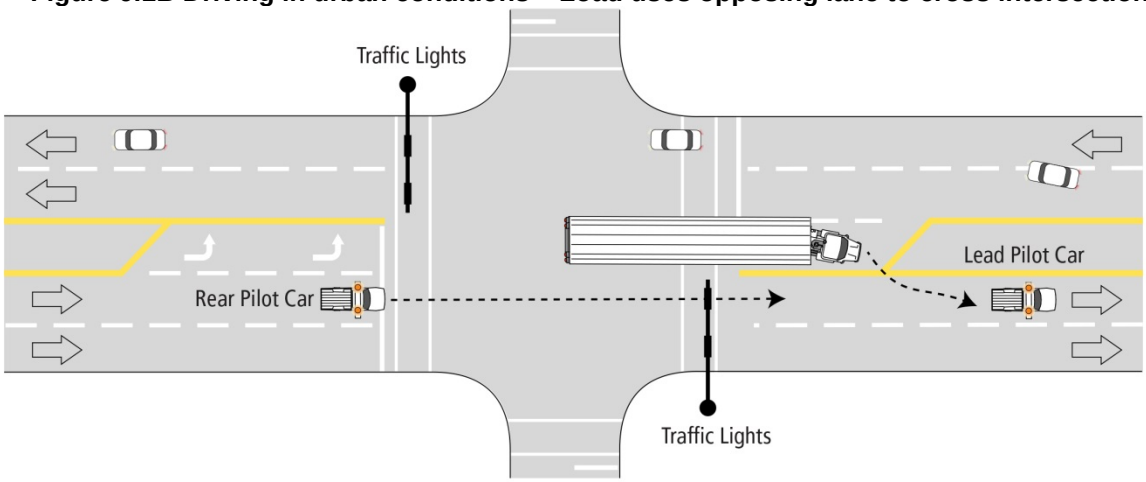


Figure 9.1C Driving in urban conditions – Load returns to travel lane

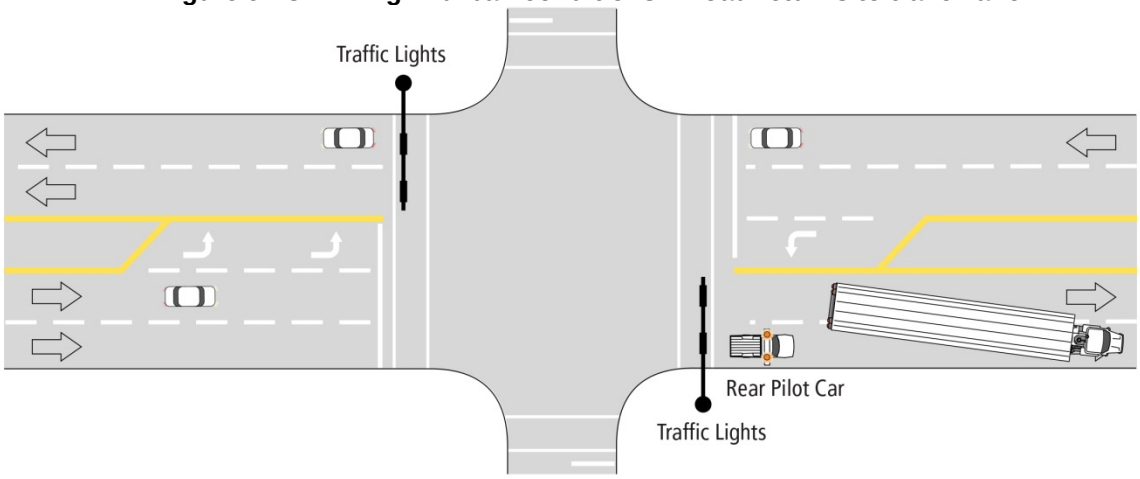
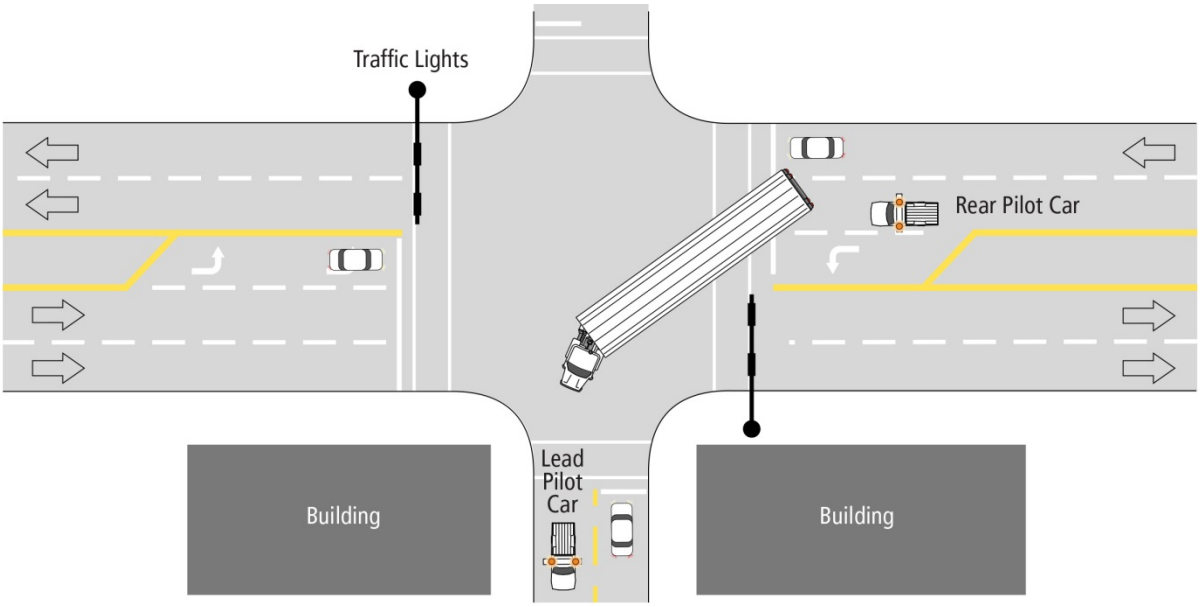


Figure 9.1D Driving in urban conditions – limited visibility at an intersection



9.2 Obeying traffic signals

Traffic signals must be obeyed when travelling a route controlled by traffic signals. Stopping for signals may cause the pilot cars and load to become separated for short durations. Pilot cars and the load driver should remain in communication and reduce speed or wait for each other, as necessary, in order to resume normal operation.

If the pilot car passes through the intersection, but the load is required to stop, the lead pilot car should pull over as soon as possible and resume travel as the load approaches.

When the rear pilot car is stopped by a red light which has been cleared by the load, the load driver (and lead pilot car) should proceed at a reduced speed until the rear pilot car has caught up.

The pilot car driver(s) must be alert and avoid sudden stops at traffic signals.